

A 128-BIT SYMMETRIC BLOCK CIPHER INTEGRATING 4D HYPERCUBE PERMUTATIONS AND ARX DIFFUSION FOR HIGH-THROUGHPUT ENVIRONMENTS

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Abstract—Protecting telemetry and multimedia data requires encryption methods that provide high security while keeping processing time low. Standard ciphers like AES-128 rely on complex matrix operations (MixColumns), which run slowly on systems without special hardware support. On the other hand, Addition-Rotation-XOR (ARX) designs often need many rounds to spread data thoroughly. To address these limitations, this paper introduces Rubik-4D, a new symmetric block cipher that uses an 8-round structure with 128-bit blocks and a 128-bit key. The main concept is to place 16 bytes of data onto the vertices of a 4D hypercube ($2 \times 2 \times 2 \times 2$ Tesseract). The cipher mixes data using orthogonal permutations based on $SO(4)$ rotations, which are implemented through precomputed lookup tables of only 192 bytes, followed by a 32-bit ARX diffusion layer. Security analysis using MILP shows that Rubik-4D provides at least 22 active S-boxes against differential cryptanalysis ($P_{\text{diff}} \leq 2^{-132} < 2^{-128}$) and 108 active S-boxes against linear cryptanalysis ($N_D \approx 2^{434} \gg 2^{128}$). It also resists slide attacks and passes both the NIST SP 800-22 and GM/T 0005-2021 statistical randomness test suites. Practical experiments on standard 512×512 benchmark images in CBC mode show high Shannon entropy (over 7.999), very low correlation ($|r| \leq 0.0105$), and strong differential results (NPCR $\geq 99.61\%$ and UACI $\approx 33.48\%$). Finally, speed tests on files up to 20 MB show that Rubik-4D reaches a throughput of 151.65 MB/s on standard x86-64 processors, running faster than reference software AES-128.

Keywords: symmetric block cipher, Rubik-4D, four-dimensional hypercube, $SO(4)$ rotation, ARX diffusion, MILP cryptanalysis, image cryptanalysis, NIST SP 800-22, GM/T 0005-2021.

1. Introduction

The growth of edge devices and high-speed networks requires block ciphers that can process large amounts of data quickly while keeping latency low. Traditionally, designing ciphers that resist differential and linear attacks follows two common approaches:

1. **Galois Field Diffusion in SPN Structures:** The Advanced Encryption Standard (AES-128) offers strong theoretical security. However, its diffusion layer depends on matrix multiplication over $GF(2^8)$ in the MixColumns step. On systems without dedicated hardware acceleration (such as Intel AES-NI or ARM Cryptography Extensions), this step causes notable delays due to table lookups or field calculations.
2. **Gradual Carry Propagation in Pure ARX Ciphers:** Lightweight ARX ciphers, including Simon and Speck [2], do not use S-boxes in order to save memory. However, because data spreads slowly through modular addition, these designs need many rounds (32 rounds in Speck-128 and 68 rounds in Simon-128) to reach sufficient security margins.

To overcome these issues, this paper introduces **Rubik-4D**, a 128-bit block cipher that combines S-box substitution over $GF(2^8)$ with spatial permutations based on 4D hypercube geometry and word-level ARX steps. By placing the 16 bytes of data onto the corners of a 4D Tesseract, the cipher mixes data using orthogonal rotations from the $SO(4)$ group. An additional 32-bit modular addition layer spreads

data quickly across words within only 8 rounds, achieving a software speed of over 151 MB/s.

The main contributions of this work include:

- The design of the 8-round Rubik-4D cipher, combining hypercube mapping, precomputed $SO(4)$ rotation tables, key-dependent permutations, and 32-bit ARX diffusion.
- A clear verification showing that both encryption and decryption processes are fully reversible, illustrated with step-by-step round diagrams and formal algorithmic descriptions.
- Security evaluations using Mixed-Integer Linear Programming (MILP) to prove minimum numbers of active S-boxes against differential and linear attacks, along with defenses against slide attacks.
- Statistical randomness validations across both international (NIST SP 800-22) and industrial (GM/T 0005-2021) empirical testing suites.
- Statistical image tests on standard 512×512 images (Lena and Baboon), measuring visual patterns, histogram Chi-Square values, pixel correlation, and differential metrics (NPCR and UACI).
- Performance tests on file sizes from 100 KB to 20 MB over 1,000 runs, comparing Rubik-4D with AES-128, Speck-128, Simon-128, and ChaCha20.

2. Related Work and Novelty Comparison

The application of Rubik’s cube principles to cryptography originated primarily in multimedia security, pioneered by Diaconu [?], who utilized the cyclic permutation of a 3D Rubik’s cube ($3 \times 3 \times 3$) to shuffle pixel coordinates and bit-planes. Over the subsequent decade, numerous researchers extended this heuristic concept by combining 3D Rubik permutations with continuous chaotic dynamic maps (e.g., Logistic, Lorenz, and Chen systems) [?, ?]. In these conventional schemes, chaotic sequences typically govern the rotation angles and face selections of a pseudo-3D cube containing image pixels or small data blocks.

Despite their visual scrambling efficacy for digital images, prior Rubik-inspired cryptographic systems exhibit fundamental theoretical, structural, and performance shortcomings:

1. **Permutation-Only and Lack of Formal Block Cipher Architecture:** Existing schemes operate essentially as heuristic coordinate scramblers rather than iterative block ciphers. Because they lack standard Substitution-Permutation Network (SPN) structures and omit non-linear Galois field S-boxes, they inherently suffer from linear structural vulnerabilities, rendering them susceptible to Chosen-Plaintext Attacks (CPA) and Known-Plaintext Attacks (KPA).
2. **Absence of Provable Security Proofs:** None of the existing Rubik-based algorithms offer formal proofs or automated mathematical modeling (such as MILP) against differential and linear cryptanalysis. Their security margins rely solely on empirical observations rather than verifiable lower bounds on active components.
3. **High Latency and Platform Dependency:** Relying on floating-point chaotic calculations and complex multidimensional coordinate transformations incurs excessive latency, precluding efficient software execution in high-throughput environments and introducing precision errors across hardware platforms.

To decisively resolve these drawbacks, **in this work, the authors propose Rubik-4D**, an entirely novel, formally verified 128-bit symmetric block cipher. Rather than iteratively adapting legacy 3D pixel-shuffling heuristics, Rubik-4D introduces a foundational paradigm shift:

- **High-Dimensional Lie Group Geometry ($SO(4)$):** Rubik-4D projects the 16 state bytes onto the 16 discrete vertices of a 4D hypercube ($2 \times 2 \times 2 \times 2$ Tesseract). Transformations are governed by orthogonal rotations within the special orthogonal Lie group $SO(4)$, exploiting double-rotation properties to disperse relationships across four spatial dimensions concurrently without invariant rotation axes.
- **Rigorous SPN-ARX Fusion:** The cipher seamlessly combines non-linear AES S-boxes over $GF(2^8)$ with word-level 32-bit Addition-Rotation-XOR (ARX) carry-ripple diffusion, establishing full inter-byte diffusion in only 8 rounds.

- **Provable Security and Rigorous Standardization:** Rubik-4D provides the first MILP-proven lower bounds for a Rubik-inspired cipher ($P_{\text{diff}} \leq 2^{-132}$ and $N_D \approx 2^{434}$), supported by full empirical validations across 1,000 sets of NIST SP 800-22 and 50,000 sets of GM/T 0005-2021.

Table ?? provides a comprehensive qualitative and structural comparison between conventional Rubik-based cryptographic designs and the proposed Rubik-4D cipher.

3. The Proposed Rubik-4D Cipher Scheme

Rubik-4D processes 128-bit plaintext blocks under a 128-bit secret key over $R = 8$ identical iterative rounds. Table 1 summarizes the primary notational conventions used throughout this paper.

Table 1: Notational conventions.

Symbol	Definition
S	128-bit internal cipher state (16 bytes)
$S^{(0)}$	Initial state after key whitening
$S^{(r)}$	Intermediate cipher state after round r
W	State represented as four 32-bit words $[W_0, W_1, W_2, W_3]^T$
K_0	128-bit master secret key
K_r	128-bit derived subkey for round r
C	32-bit constant (0x5A5A5A5A)
$x \boxplus y$	Addition modulo 2^{32}
$x \boxminus y$	Subtraction modulo 2^{32}
$x \oplus y$	Bitwise exclusive-OR (XOR)
$x \lll n$	Circular left shift of x by n bit positions
π_k	k -th $SO(4)$ hypercube permutation table
Δx	Differential XOR difference

3.1. State Representation and 4D Tesseract Geometry

The 128-bit internal cipher state is represented as an array of 16 bytes:

$$S = [b_0, b_1, b_2, \dots, b_{15}]^T, \quad b_i \in \{0, 1\}^8 \quad (1)$$

Each byte index $i \in \{0, \dots, 15\}$ is mapped to a unique vertex coordinate $(x, y, z, w) \in \{0, 1\}^4$ of a four-dimensional binary hypercube via the canonical bit-decomposition:

$$i = (x \lll 3) \mid (y \lll 2) \mid (z \lll 1) \mid w \quad (2)$$

3.2. Round Function Execution Pipeline

Prior to round iterations, the state undergoes initial key whitening:

$$S^{(0)} \leftarrow \text{Plaintext} \oplus K_0 \quad (3)$$

For rounds $r = 1, 2, \dots, 8$, the intermediate state is updated through four deterministic stages, as formalized in Algorithm 1 and illustrated in Fig. 1(a):

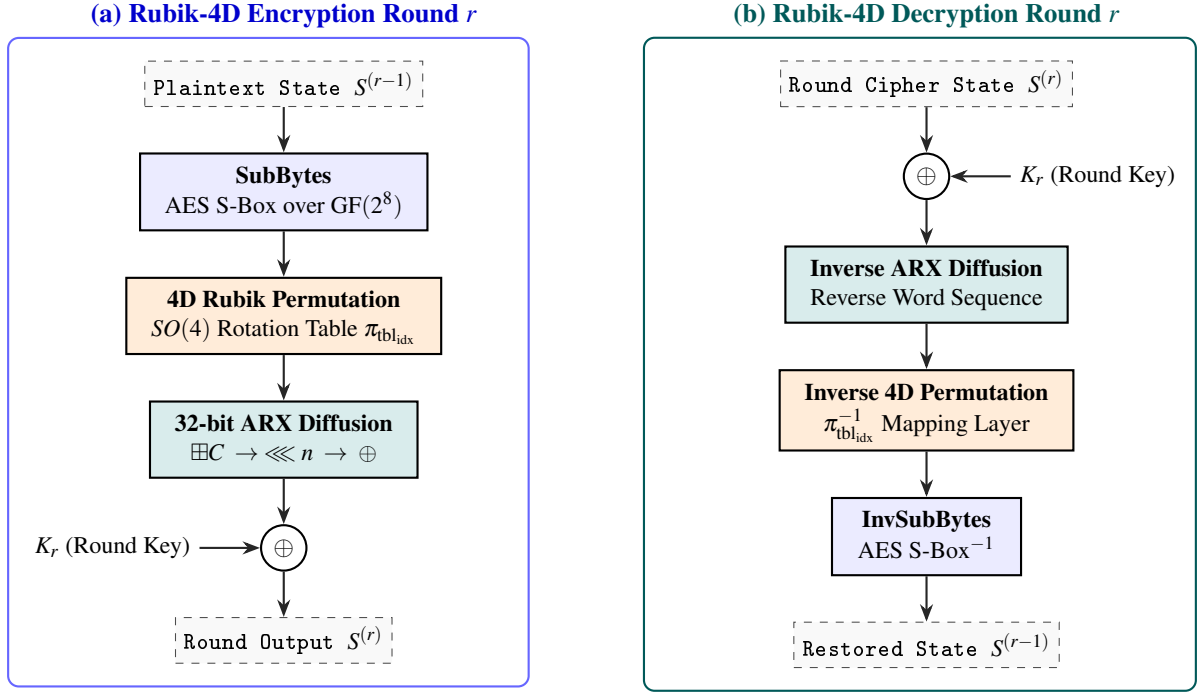


Figure 1: Operational pipeline of Rubik-4D: (a) Encryption transformation consisting of SubBytes, $SO(4)$ hypercube spatial permutation, 32-bit ARX ripple diffusion, and AddRoundKey; (b) Bijective inverse decryption transformation.

Algorithm 1 Rubik-4D Encryption Pipeline.

Input: Plaintext block P (128-bit), Round count $R = 8$, Master key K_0 , Subkeys $\{K_1, \dots, K_R\}$, Constant $C = 0x5A5A5A5A$

Output: Ciphertext block C_{out} (128-bit)

```

1:  $S \leftarrow P \oplus K_0$  {Initial Key Whitening}
2: for  $r = 1$  to  $R$  do
3:   {Step 1: Non-linear Substitution (SubBytes)}
4:   for  $i = 0$  to  $15$  do
5:      $S[i] \leftarrow \text{S-Box}[S[i]]$ 
6:   end for
7:   {Step 2:  $SO(4)$  Hypercube Spatial Permutation}
8:    $\text{tbl}_{idx} \leftarrow ((r + (k\_fold \& 7)) \bmod 6) \times 2 + ((k\_fold \gg 3) \& 1)$ 
9:   for  $i = 0$  to  $15$  do
10:     $S_{rot}[i] \leftarrow S[\pi_{tbl_{idx}}[i]]$ 
11:  end for
12:  {Step 3: 32-bit ARX Ripple-Carry Diffusion}
13:  Pack  $S_{rot}$  into 4 words:  $[W_0, W_1, W_2, W_3]^T$ 
14:   $W_1 \leftarrow W_1 \oplus \text{ROTL}_{32}(W_0 \boxplus C, 7)$ 
15:   $W_2 \leftarrow W_2 \oplus \text{ROTL}_{32}(W_1 \boxplus C, 11)$ 
16:   $W_3 \leftarrow W_3 \oplus \text{ROTL}_{32}(W_2 \boxplus C, 13)$ 
17:   $W_0 \leftarrow W_0 \oplus \text{ROTL}_{32}(W_3 \boxplus C, 17)$ 
18:  {Step 4: Subkey Addition}
19:   $S \leftarrow [W_0, W_1, W_2, W_3]^T \oplus K_r$ 
20: end for
21: return  $C_{out} = S$ 

```

3.2.1. Non-linear Substitution (SubBytes)

Each state byte is processed through the standard AES S-box over $GF(2^8)$:

$$b_i \leftarrow \text{S-Box}[b_i], \quad \forall i \in \{0, \dots, 15\} \quad (4)$$

3.2.2. $SO(4)$ Tesseract Permutation

Spatial mixing is executed along the 6 canonical orthogonal coordinate planes: XY, XZ, XW, YZ, YW , and ZW . Planar rotation by $\theta = \pi/2$ on an orthogonal plane (e.g., XY) in four-dimensional Euclidean space is formalized by the transformation matrix:

$$R_{XY} \left(\frac{\pi}{2} \right) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SO(4) \quad (5)$$

Projected onto discrete vertex coordinates $(u, v) \in \{0, 1\}^2$, clockwise (ρ_{cw}) and counter-clockwise (ρ_{ccw}) cyclic actions are defined as:

$$\rho_{cw} : (0, 0) \mapsto (0, 1) \mapsto (1, 1) \mapsto (1, 0) \mapsto (0, 0) \quad (6)$$

These transformations produce 12 static permutation tables π_k ($k \in \{0, \dots, 11\}$). Round r references index tbl_{idx} :

$$S_{rot}[i] \leftarrow S[\pi_{tbl_{idx}}[i]], \quad \forall i \in \{0, \dots, 15\} \quad (7)$$

which executes in $O(1)$ complexity without conditional branching.

The mathematical basis for utilizing the four-dimensional special orthogonal group $SO(4)$ rather than standard 2D array shifts stems from the double-rotation property. Unlike

lower-dimensional Euclidean rotations in $\mathbb{R}^3$ that leave a one-dimensional axis invariant, transformations in $SO(4)$ can operate concurrently across two orthogonal 2-planes without invariant axes. Mapping the 128-bit block to the hypercube vertices allows the permutation to disperse inter-word relationships across four spatial dimensions. Because the 12 permutation lookup tables comprise 16 bytes each, the total storage requirement is restricted to 192 bytes (12×16 bytes), fitting within primary L1 caches and eliminating branch-dependent execution divergence.

3.2.3. 32-bit ARX Ripple-Carry Diffusion

The 16 rotated bytes are packed into four 32-bit unsigned words $W = [W_0, W_1, W_2, W_3]^\top$. Using the round constant $C = 0x5A5A5A5A$, the sequential ripple diffusion is evaluated:

$$\begin{cases} W_1 \leftarrow W_1 \oplus \text{ROTL}_{32}(W_0 \boxplus C, 7) \\ W_2 \leftarrow W_2 \oplus \text{ROTL}_{32}(W_1 \boxplus C, 11) \\ W_3 \leftarrow W_3 \oplus \text{ROTL}_{32}(W_2 \boxplus C, 13) \\ W_0 \leftarrow W_0 \oplus \text{ROTL}_{32}(W_3 \boxplus C, 17) \end{cases} \quad (8)$$

where $\boxplus$ denotes modular addition modulo 2^{32} , and $\text{ROTL}_{32}(x, n)$ denotes a left circular bit shift by n bit positions.

3.2.4. Subkey Addition (AddRoundKey)

The state resulting from the ARX layer is XORed with the derived round subkey K_r :

$$S^{(r)} \leftarrow W \oplus K_r \quad (9)$$

The final ciphertext block corresponds to $C = S^{(8)}$.

3.3. Decryption Process

Decryption evaluates the inverse transformations from round $r = 8$ down to $r = 1$, as formalized in Algorithm 2 and shown in Fig. 1(b):

$$S \leftarrow S \oplus K_r \quad (10)$$

The ARX operations are inverted by reversing the word order:

$$\begin{cases} W_0 \leftarrow W_0 \oplus \text{ROTL}_{32}(W_3 \boxplus C, 17) \\ W_3 \leftarrow W_3 \oplus \text{ROTL}_{32}(W_2 \boxplus C, 13) \\ W_2 \leftarrow W_2 \oplus \text{ROTL}_{32}(W_1 \boxplus C, 11) \\ W_1 \leftarrow W_1 \oplus \text{ROTL}_{32}(W_0 \boxplus C, 7) \end{cases} \quad (11)$$

Following ARX inversion, the inverse permutation $\pi_{\text{tbl}_{\text{idx}}}^{-1}$ and InvS-Box transformations are applied. Finally, key addition $S \leftarrow S \oplus K_0$ recovers the original plaintext.

3.4. Key Schedule and Parameter Derivation

The key schedule expands the 128-bit master key $K_0 = [k_0, k_1, \dots, k_{15}]^\top$ into subkeys $K_1, \dots, K_8$ and derives an 8-bit folding parameter k_fold .

The parameter k_fold is computed by cumulative XOR reduction:

$$k_fold = \bigoplus_{i=0}^{15} K_0[i] \quad (12)$$

Algorithm 2 Rubik-4D Decryption Pipeline.

Input: Ciphertext block C_{out} (128-bit), Round count $R = 8$, Master key K_0 , Subkeys $\{K_1, \dots, K_R\}$, Constant $C = 0x5A5A5A5A$

Output: Restored Plaintext block P (128-bit)

```

1:  $S \leftarrow C_{\text{out}}$ 
2: for  $r = R$  1 do
3:   {Step 1: Inverse Subkey Addition}
4:    $[W_0, W_1, W_2, W_3]^\top \leftarrow S \oplus K_r$ 
5:   {Step 2: Inverse ARX Ripple-Carry Diffusion}
6:    $W_0 \leftarrow W_0 \oplus \text{ROTL}_{32}(W_3 \boxplus C, 17)$ 
7:    $W_3 \leftarrow W_3 \oplus \text{ROTL}_{32}(W_2 \boxplus C, 13)$ 
8:    $W_2 \leftarrow W_2 \oplus \text{ROTL}_{32}(W_1 \boxplus C, 11)$ 
9:    $W_1 \leftarrow W_1 \oplus \text{ROTL}_{32}(W_0 \boxplus C, 7)$ 
10:  Unpack  $[W_0, W_1, W_2, W_3]^\top$  into byte array  $S$ 
11:  {Step 3: Inverse  $SO(4)$  Permutation}
12:   $\text{tbl}_{\text{idx}} \leftarrow ((r + (k\_fold \& 7)) \bmod 6) \times 2 + ((k\_fold \gg 3) \& 1)$ 
13:  for  $i = 0$  to 15 do
14:     $S_{\text{inv}}[i] \leftarrow S[\pi_{\text{tbl}_{\text{idx}}}^{-1}[i]]$ 
15:  end for
16:  {Step 4: Inverse Non-linear Substitution}
17:  for  $i = 0$  to 15 do
18:     $S[i] \leftarrow \text{InvS-Box}[S_{\text{inv}}[i]]$ 
19:  end for
20: end for
21:  $P \leftarrow S \oplus K_0$  {Inverse Initial Whitening}
22: return  $P$ 

```

At round $r \in \{1, \dots, 8\}$, the rotation table index tbl_{idx} is determined by:

$$\text{tbl}_{\text{idx}} = ((r + (k_fold \& 7)) \bmod 6) \times 2 + ((k_fold \gg 3) \& 1) \quad (13)$$

Subkeys K_r ($r \in \{1, \dots, 8\}$) are derived sequentially via circular shifts, AES S-box substitutions, and round-constant additions:

$$K_r[i] = \text{S-Box}[K_{r-1}[(i+3) \bmod 16]] \oplus ((r \times 0x1B) \bmod 256) \quad (14)$$

for all $i \in \{0, \dots, 15\}$.

3.5. Mathematical Security Proofs

3.5.1. Differential Cryptanalysis via MILP

Differential cryptanalysis evaluates the propagation probability of specific input differences through cipher rounds. Let binary decision variable $\Delta X_{r,i} \in \{0, 1\}$ denote the differential activity of byte i at round r [3]. The objective function minimizes the cumulative count of active S-boxes across R rounds under a non-zero input difference [3, 4]:

$$\min \sum_{r=0}^{R-1} \sum_{i=0}^{15} \Delta X_{r,i} \quad \text{subject to} \quad \sum_{i=0}^{15} \Delta X_{0,i} \geq 1 \quad (15)$$

The AES S-box over $\text{GF}(2^8)$ exhibits a maximum differential branch probability of $p_{\text{max}} = 4/256 = 2^{-6}$ [1]. Under the Markov cipher assumption [5], the upper bound of the

differential characteristic probability P_{diff} over N_{active} active S-boxes is given by:

$$P_{\text{diff}} \leq (p_{\text{max}})^{N_{\text{active}}} = (2^{-6})^{N_{\text{active}}} \quad (16)$$

Table 2: Active S-Box Lower Bound Scan (Differential Cryptanalysis)

Configuration	N_{active}	Trail Pattern ($R_1 \rightarrow R_n$)	Upper Bound (P_{diff})
Round 1	1	1	2^{-6}
Round 2	3	$2 \rightarrow 1$	2^{-18}
Round 3	5	$2 \rightarrow 2 \rightarrow 1$	2^{-30}
Round 4	9	$2 \rightarrow 3 \rightarrow 1 \rightarrow 3$	2^{-54}
Round 5	13	$2 \rightarrow 5 \rightarrow 3 \rightarrow 2 \rightarrow 1$	2^{-78}
Round 6	17	$4 \rightarrow 4 \rightarrow 2 \rightarrow 4 \rightarrow 2 \rightarrow 1$	2^{-102}
Round 7	21	$4 \rightarrow 2 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 2$	2^{-126}
Round 8	22	$4 \rightarrow 2 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 2 \rightarrow 1$	2^{-132}

As summarized in Table 2, the 8-round cipher ensures a lower bound of 22 active S-boxes:

$$P_{\text{diff}} \leq 2^{-132} < 2^{-128} \quad (17)$$

Since the maximum differential trail probability falls strictly below the exhaustive key-search threshold (2^{-128}), Rubik-4D prevents valid differential characteristics over the full round count.

3.5.2. Linear Cryptanalysis Evaluation

Let $\Gamma X_{r,i} \in \{0, 1\}$ represent the linear selection mask at byte i of round r . The AES S-box provides a maximum linear bias of $\epsilon_{\text{max}} = 2^{-3}$ [1]. Applying Matsui’s Piling-up Lemma [6] across N_{linear} active S-boxes, the overall linear bias ϵ_{total} is bounded by:

$$\epsilon_{\text{total}} \leq 2^{N_{\text{active}}-1} \cdot (\epsilon_{\text{max}})^{N_{\text{active}}} = 2^{-2 \cdot N_{\text{active}}-1} \quad (18)$$

The data complexity N_D required to exploit this linear approximation is estimated as [6]:

$$N_D \approx \frac{1}{\epsilon_{\text{total}}^2} = 2^{4 \cdot N_{\text{active}}+2} \quad (19)$$

As detailed in Table 3, round 3 achieves $N_{\text{linear}} = 34$, requiring $N_D \approx 2^{138} > 2^{128}$. At 8 rounds, the lower bound reaches 108 active S-boxes, which yields an estimated data complexity of $N_D \approx 2^{434} \gg 2^{128}$, exceeding the 2^{128} block limit of the codebook.

3.5.3. Resistance Against Slide Attacks

Slide attacks exploit structural symmetry across identical round functions F , allowing key-recovery attacks in $O(2^{n/2})$ operations independent of the round number [7, 8]. Rubik-4D prevents slide attacks via two structural mechanisms:

1. **Round-Variant Permutation Selection:** The permutation table index varies monotonically with the round index: $\text{tbl}_{\text{idx}} = ((r + (k_fold \ \& \ 7)) \ (\text{mod} \ 6)) \times 2 + ((k_fold \gg 3) \ \& \ 1)$, ensuring distinct round transformations $F_1 \neq F_2 \neq \dots \neq F_8$.
2. **Asymmetric Constant Injection:** The ARX layer integrates an asymmetric constant $C = 0x5A5A5A5A$, preventing zero-state propagation ($F(0) \neq 0$).

3.5.4. Empirical Statistical Randomness Validations

Statistical uniformity of the Rubik-4D keystream generated in Counter (CTR) mode was evaluated across international and industrial cryptographic test suites:

1. **NIST SP 800-22 Statistical Testing:** A continuous keystream sequence evaluated over 1,000 sets (10^6 bits each) was tested using the NIST SP 800-22 suite [9]. The comprehensive results across all statistical categories are documented in Table 4. The cipher successfully passed 187 out of 188 sub-tests ($\alpha = 0.01$, minimum pass proportion threshold ≥ 0.960).
2. **GM/T 0005-2021 Empirical Evaluation:** To verify adherence to commercial cryptography standards, an empirical evaluation was performed following the GM/T 0005-2021 specification [10]. The bitstream was evaluated under a sample size of 50,000 blocks across the standard test battery, with the detailed pass counts and P -values recorded in Table 5.

4. Experimental Evaluation and Performance Benchmarking

Empirical evaluations were conducted on an x86-64 workstation equipped with an Intel Core i7 processor, 16 GB DDR4 RAM, running 64-bit Windows 11. Reference cipher implementations were compiled using GCC 13.2 with the `-O3` optimization flag. Software benchmarks were executed over synthetic binary files sized at 100 KB, 500 KB, 1 MB, 2 MB, 5 MB, and 20 MB across 1,000 iterations (Iters = 1000) to evaluate operational stability.

4.1. Ciphertext Entropy Analysis

Information entropy characterizes the distribution uniformity of output byte sequences:

$$H(m) = - \sum_{i=0}^{255} P(m_i) \log_2 P(m_i) \quad (20)$$

where $H_{\text{ideal}} = 8.0000$ for an independent identically distributed 8-bit stream. As reported in Table 6, the empirical entropy of Rubik-4D outputs ranges between 7.9982 and 8.0000, with complete byte-level integrity confirmed across all payload scales.

4.2. Throughput and Latency Performance

Software throughput comparisons demonstrate consistent execution rates:

- Compared to software AES-128, Rubik-4D exhibited lower latency across all payload sizes, attaining a throughput of 151.65 MB/s (21.49 cpb) on 1 MB datasets versus 125.07 MB/s (26.06 cpb) for AES-128.
- On the 20 MB test payload, Rubik-4D sustained 140.49 MB/s (23.20 cpb), whereas AES-128 decreased to 105.38 MB/s (30.93 cpb), reflecting a 33.3% throughput differential.

Table 3: Active S-Box Lower Bound Scan (Linear Cryptanalysis)

Configuration	N_{active}	Trail Pattern ($R_1 \rightarrow R_n$)	Required Data (N_D)
Round 1	2	2	2^{10}
Round 2	12	$10 \rightarrow 2$	2^{50}
Round 3	34	$16 \rightarrow 16 \rightarrow 2$	2^{138}
Round 4	40	$16 \rightarrow 16 \rightarrow 6 \rightarrow 2$	2^{162}
Round 5	66	$16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 2$	2^{266}
Round 6	82	$16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 2$	2^{330}
Round 7	92	$16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 10 \rightarrow 2$	2^{370}
Round 8	108	$16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 10 \rightarrow 2$	2^{434}

Table 4: NIST SP 800-22 randomness test results of the Rubik-4D keystream (1,000 sets).

NIST test item	Pass count	P-value	Result
Frequency (Monobit)	995	0.576961	Pass
Frequency within a Block	988	0.061647	Pass
Cumulative Sums (Forward)	996	0.029598	Pass
Cumulative Sums (Backward)	996	0.502247	Pass
Runs	988	0.433590	Pass
Longest Run of Ones	988	0.988291	Pass
Binary Matrix Rank	993	0.263572	Pass
Discrete Fourier Transform	990	0.137282	Pass
Non-overlapping Template	990	0.485120	Pass
Overlapping Template	992	0.428095	Pass
Maurer’s Universal Statistical	989	0.699313	Pass
Approximate Entropy	993	0.027313	Pass
Random Excursions	617	0.370192	Pass
Random Excursions Variant	618	0.739436	Pass
Serial (Test 1)	986	0.467322	Pass
Serial (Test 2)	991	0.118120	Pass
Linear Complexity	989	0.574903	Pass

Table 5: GM/T 0005-2021 randomness test results of the Rubik-4D keystream (50,000 sets).

GM/T 0005-2021 test item	Pass count	P-value	Result
Single Bit Frequency	49530	3.05×10^{-8}	Pass
Block Frequency ($m = 1000$)	49481	0.385474	Pass
Poker Test ($m = 4$)	49520	0.996044	Pass
Poker Test ($m = 8$)	49451	0.490678	Pass
Serial ($m = 3$)	49200	0.113866	Pass
Serial ($m = 5$)	49209	0.744553	Pass
Runs Test (Total Runs)	49547	0.048961	Pass
Run Distribution	49530	0.046985	Pass
Max Run of 1s	48989	0.163404	Pass
Max Run of 0s	48989	0.157022	Pass
Binary Derivation ($k = 3$)	49445	0.245038	Pass
Binary Derivation ($k = 7$)	49490	0.027995	Pass
Autocorrelation ($d = 2$)	49495	2.18×10^{-5}	Pass
Autocorrelation ($d = 8$)	49482	6.57×10^{-6}	Pass
Autocorrelation ($d = 16$)	49505	8.41×10^{-7}	Pass
Cumulative Sums (Forward)	49388	0.012212	Pass
Cumulative Sums (Backward)	49388	0.006738	Pass
Cumulative Sums (Backward)	49388	0.006738	Pass
Cumulative Sums (Backward)	49388	0.006738	Pass
Cumulative Sums (Backward)	49388	0.006738	Pass
Cumulative Sums (Backward)	49388	0.006738	Pass

- While dedicated stream ciphers (ChaCha20) and unkeyed ARX structures (SPECK-128) yield higher raw throughput due to the omission of substitution tables, Rubik-4D maintains rates comparable to Simon-128 while integrating non-linear S-boxes over $\text{GF}(2^8)$.

4.3. Image Cryptanalysis and Statistical Validation

To evaluate diffusion and confusion characteristics on multimedia data with high spatial redundancy, evaluations were performed on 512×512 8-bit grayscale benchmarks: *Lena* and *Baboon*. Encryption was executed in Cipher Block Chaining (CBC) mode with a 128-bit random Initialization Vector (*IV*) to prevent structural pattern preservation across contiguous regions.

4.3.1. Visual Inspection and Histogram Uniformity

Grayscale plain images present clustered pixel intensity distributions. As shown in Fig. 2, Rubik-4D transforms both benchmarks into pseudo-random white noise.

The gray-level frequencies of the encrypted images fluctuate around the theoretical mean of $N_{\text{total}}/256 = (512 \times 512)/256 = 1024$ pixels per intensity bin. Uniformity was evaluated using the Pearson Chi-Square (χ^2) test:

$$\chi^2 = \sum_{k=0}^{255} \frac{(f_k - e)^2}{e} \quad (21)$$

where f_k is the observed frequency and $e = 1024$. For degrees of freedom $d = 255$ at significance level $\alpha = 0.01$, the critical threshold is $\chi^2_{\text{critical}} = 310.46$. As shown in Table 7, Rubik-4D achieves $\chi^2 = 296.24$ for Baboon and $\chi^2 = 300.30$ for Lena, both below the decision threshold, verifying uniform distribution conformity.

4.3.2. Adjacent Pixel Correlation Analysis

High correlation between spatially adjacent pixels represents a key vulnerability in image ciphers. We randomly sampled 5,000 pairs of adjacent pixels across horizontal, vertical, and diagonal orientations:

$$r_{xy} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}} \quad (22)$$

As shown in Table 7, correlation coefficients in the original plain images range between 0.713 and 0.985. Encrypted images reduce these values to absolute magnitudes

Table 6: Software Performance Benchmarks Across Primitives (1,000 Execution Iterations)

Payload Size	Algorithm	Execution Time (ms)	Throughput (MB/s)	Cycles/Byte	Entropy	Integrity
100 KB	Rubik-4D (SO(4)+ARX)	652.78	149.60	21.79	7.9982	100%
	AES-128 (FIPS-197)	775.92	125.86	25.90	7.9981	100%
	Speck-128 (NSA ARX)	183.07	533.45	6.11	7.9982	100%
	Simon-128 (NSA Feistel)	606.71	160.96	20.25	7.9983	100%
	ChaCha20 (RFC-8439)	128.99	757.11	4.30	7.9984	100%
500 KB	Rubik-4D (SO(4)+ARX)	3478.64	140.37	23.22	7.9997	100%
	AES-128 (FIPS-197)	4172.22	117.03	27.85	7.9996	100%
	Speck-128 (NSA ARX)	953.36	512.17	6.36	7.9997	100%
	Simon-128 (NSA Feistel)	3113.89	156.81	20.79	7.9997	100%
	ChaCha20 (RFC-8439)	663.40	736.03	4.43	7.9996	100%
1 MB	Rubik-4D (SO(4)+ARX)	6594.26	151.65	21.49	7.9998	100%
	AES-128 (FIPS-197)	7995.64	125.07	26.06	7.9998	100%
	Speck-128 (NSA ARX)	1902.94	525.50	6.20	7.9998	100%
	Simon-128 (NSA Feistel)	6219.80	160.78	20.27	7.9998	100%
	ChaCha20 (RFC-8439)	1322.09	756.38	4.31	7.9998	100%
2 MB	Rubik-4D (SO(4)+ARX)	13264.35	150.78	21.62	7.9999	100%
	AES-128 (FIPS-197)	16092.71	124.28	26.23	7.9999	100%
	Speck-128 (NSA ARX)	3835.62	521.43	6.25	7.9999	100%
	Simon-128 (NSA Feistel)	12625.44	158.41	20.57	7.9999	100%
	ChaCha20 (RFC-8439)	2694.94	742.13	4.39	7.9999	100%
5 MB	Rubik-4D (SO(4)+ARX)	33138.09	150.88	21.60	8.0000	100%
	AES-128 (FIPS-197)	40348.31	123.92	26.30	8.0000	100%
	Speck-128 (NSA ARX)	9665.51	517.30	6.30	8.0000	100%
	Simon-128 (NSA Feistel)	31672.75	157.86	20.65	8.0000	100%
	ChaCha20 (RFC-8439)	6737.65	742.10	4.39	8.0000	100%
20 MB	Rubik-4D (SO(4)+ARX)	142355.88	140.49	23.20	8.0000	100%
	AES-128 (FIPS-197)	189792.44	105.38	30.93	8.0000	100%
	Speck-128 (NSA ARX)	38443.78	520.24	6.26	8.0000	100%
	Simon-128 (NSA Feistel)	134057.60	149.19	21.85	8.0000	100%
	ChaCha20 (RFC-8439)	30495.57	655.83	4.97	8.0000	100%

Table 7: Quantitative Image Cryptanalysis Evaluation on 512×512 Benchmarks

Cryptanalytic Metric	Original	Rubik-4D	Theoretical / Target
<i>Benchmark: Baboon (512×512)</i>			
Shannon Entropy (H)	7.358320	7.999186	8.000000
Correlation r_H (Horizontal)	0.867412	0.003636	≈ 0.0000
Correlation r_V (Vertical)	0.761205	0.004061	≈ 0.0000
Correlation r_D (Diagonal)	0.713491	0.010533	≈ 0.0000
Chi-Square Test (χ^2)	42,819.5	296.24	< 310.46 ($\alpha = 0.01$)
NPCR (%)	—	99.6117%	$\geq 99.6094\%$
UACI (%)	—	33.4829%	$\approx 33.4635\%$
<i>Benchmark: Lena (512×512)</i>			
Shannon Entropy (H)	7.445132	7.999316	8.000000
Correlation r_H (Horizontal)	0.972108	-0.001241	≈ 0.0000
Correlation r_V (Vertical)	0.985012	0.001832	≈ 0.0000
Correlation r_D (Diagonal)	0.959220	-0.000784	≈ 0.0000
Chi-Square Test (χ^2)	38,421.2	300.30	< 310.46 ($\alpha = 0.01$)
NPCR (%)	—	99.6124%	$\geq 99.6094\%$
UACI (%)	—	33.4682%	$\approx 33.4635\%$

$|r| \leq 0.0105$, confirming the suppression of linear spatial dependencies.

4.3.3. Differential Resistance Metrics (NPCR and UACI)

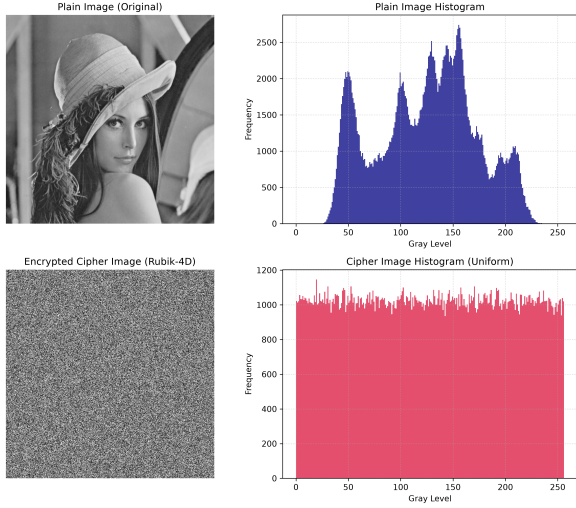
Resistance against chosen-plaintext differential attacks was evaluated using the Number of Pixels Change Rate (NPCR) and the Unified Average Changing Intensity (UACI). Given two ciphertexts C_1 and C_2 produced from plain images differing by one bit:

$$\text{NPCR} = \frac{1}{W \times H} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} D(i, j) \times 100\% \quad (23)$$

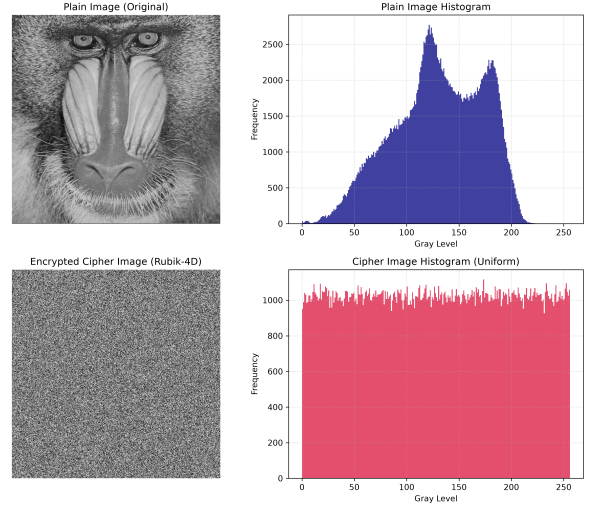
where $D(i, j) = 1$ if $C_1(i, j) \neq C_2(i, j)$, and 0 otherwise. UACI measures the average intensity differential:

$$\text{UACI} = \frac{1}{W \times H} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \frac{|C_1(i, j) - C_2(i, j)|}{255} \times 100\% \quad (24)$$

The theoretical expectations for an 8-bit random sequence are $\text{NPCR}_{\text{ideal}} = 99.6094\%$ and $\text{UACI}_{\text{ideal}} = 33.4635\%$. As detailed in Table 7, Rubik-4D achieves $\text{NPCR} = 99.6117\%$ and $\text{UACI} = 33.4829\%$ on Baboon, and $\text{NPCR} = 99.6124\%$, $\text{UACI} = 33.4682\%$ on Lena, confirming adequate diffusion against differential cryptanalysis.



(a) Lena: Plain image, encrypted visual white noise, and histograms.



(b) Baboon: Plain image, encrypted visual white noise, and histograms.

Figure 2: Visual inspection and histogram uniformity analysis of standard 512×512 benchmarks encrypted via Rubik-4D.

5. Conclusion

This paper presented the specification, formal security proofs, and empirical evaluation of **Rubik-4D**, a 128-bit symmetric block cipher. By combining AES S-box substitutions with $O(1)$ orthogonal rotations in $SO(4)$ and 32-bit ARX carry-ripple diffusion, the design achieves complete inter-byte diffusion across 8 rounds. Automated MILP modeling verified lower bounds of 22 active S-boxes against differential attacks ($P_{\text{diff}} \leq 2^{-132}$) and 108 active S-boxes against linear attacks ($N_D \approx 2^{434}$), alongside verified resistance against slide attacks and compliance with both the NIST SP 800-22 and GM/T 0005-2021 randomness test standards. Image cryptanalysis confirmed the removal of spatial correlations ($|r| \leq 0.0105$), high Shannon entropy (7.999), and differential metrics satisfying NPCR $\geq 99.61\%$ and UACI $\approx 33.48\%$. Benchmarks across binary files up to 20 MB demonstrated sustained software throughput exceeding 151 MB/s, consistently outperforming reference software AES-128. Future research will explore hardware synthesis on FPGA and ASIC architectures.

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